

Physical Chemistry (I)

Lecture 22

Electrolytes



In the Last Class

- Chemical reactions and equilibria
 - Reaction quotient and equilibrium constant
 - Activity approximations
 - Gaseous reactions
- Le Chatelier's principle

Last class

Chemical Reaction

For an arbitrary reaction,

$$\sum_{\text{reactant } i} \nu_i \{i\} \rightleftharpoons \sum_{\text{product } j} \nu_j \{j\}$$



$$\nu_A = 2, \nu_B = 1, \nu_C = 3, \nu_D = 1$$

Now,

$$dG = \sum_{\text{product } j} \mu_j dn_j + \sum_{\text{reactant } i} \mu_i dn_i$$



Last class

Extent of Reaction

$$dG = \sum_{\text{product } j} \mu_j dn_j + \sum_{\text{reactant } i} \mu_i dn_i$$

$$dG = \left(\sum_{\text{product } j} \mu_j \nu_j - \sum_{\text{reactant } i} \mu_i \nu_i \right) d\xi$$

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi} \right)_{p,T} = \sum_{\text{product } j} \mu_j \nu_j - \sum_{\text{reactant } i} \mu_i \nu_i$$

Activity

$$\Delta_r G = \sum_{\text{product } j} \nu_j \mu_j - \sum_{\text{reactant } i} \nu_i \mu_i$$

For each species x ,

$$\nu_x \mu_x = \nu_x (\mu_x^\ominus + RT \ln a_x) = \nu_x \mu_x^\ominus + RT \ln(a_x^{\nu_x})$$

and one can show

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \left(\prod_{\text{product } j} a_j^{\nu_j} / \prod_{\text{reactant } i} a_i^{\nu_i} \right)$$

Last class

Q and K

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q$$

where the **reaction quotient** Q is defined as

$$Q = \frac{\prod_{\text{product } j} a_j^{v_j}}{\prod_{\text{reactant } i} a_i^{v_i}}$$

At equilibrium, $\Delta_r G = 0$

$$0 = \Delta_r G^\ominus + RT \ln Q_{\text{eq}} = \Delta_r G^\ominus + RT \ln K$$

where the **equilibrium constant** K is Q at equilibrium



Approximation for Activity

State	Measure	Approximation for a_j	Definition
Solute	molality	$b_j/b^\ominus$	$b^\ominus = 1 \text{ mol kg}^{-1}$
	molar concentration	$[J]/c^\ominus$	$c^\ominus = 1 \text{ mol dm}^{-3}$
Gas phase	partial pressure	$p_j/p^\ominus$	$p^\ominus = 1 \text{ bar}$
Pure solid, liquid		1 (exact)	

Equilibrium Constants

For gaseous reactions (assuming perfect gases),

$$K_p = \frac{\prod_{\text{product } j} (p_j/p^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (p_i/p^\ominus)^{\nu_i}}$$

$$K_c = \frac{\prod_{\text{product } j} ([j]/c^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} ([i]/c^\ominus)^{\nu_i}}$$

$$K_p = K_c \left(\frac{c^\ominus RT}{p^\ominus} \right)^{\Delta \nu}$$

Theoretical Estimation

For a perfect-gas reaction,

$$\mu_x = -RT \ln \left(\frac{q_x}{N_x} \right)$$

and one can show

$$K_c = \frac{\prod_{\text{product } j} ([j]/c^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} ([i]/c^\ominus)^{\nu_i}} = \frac{\prod_{\text{product } j} \{q_j/(N_A V c^\ominus)\}^{\nu_j}}{\prod_{\text{reactant } i} \{q_i/(N_A V c^\ominus)\}^{\nu_i}}$$

Le Chatelier's Principle

When subjected to a disturbance,
a system at equilibrium tends to respond in a way
that **minimizes the effect of the disturbance**.

Disturbances to consider:

- Amount
- Volume & pressure (for a gaseous reaction)
- Temperature

Last class

van't Hoff Equation

$$\ln K = -\Delta_r G^\ominus / RT$$

Using the Gibbs-Helmholtz equation,

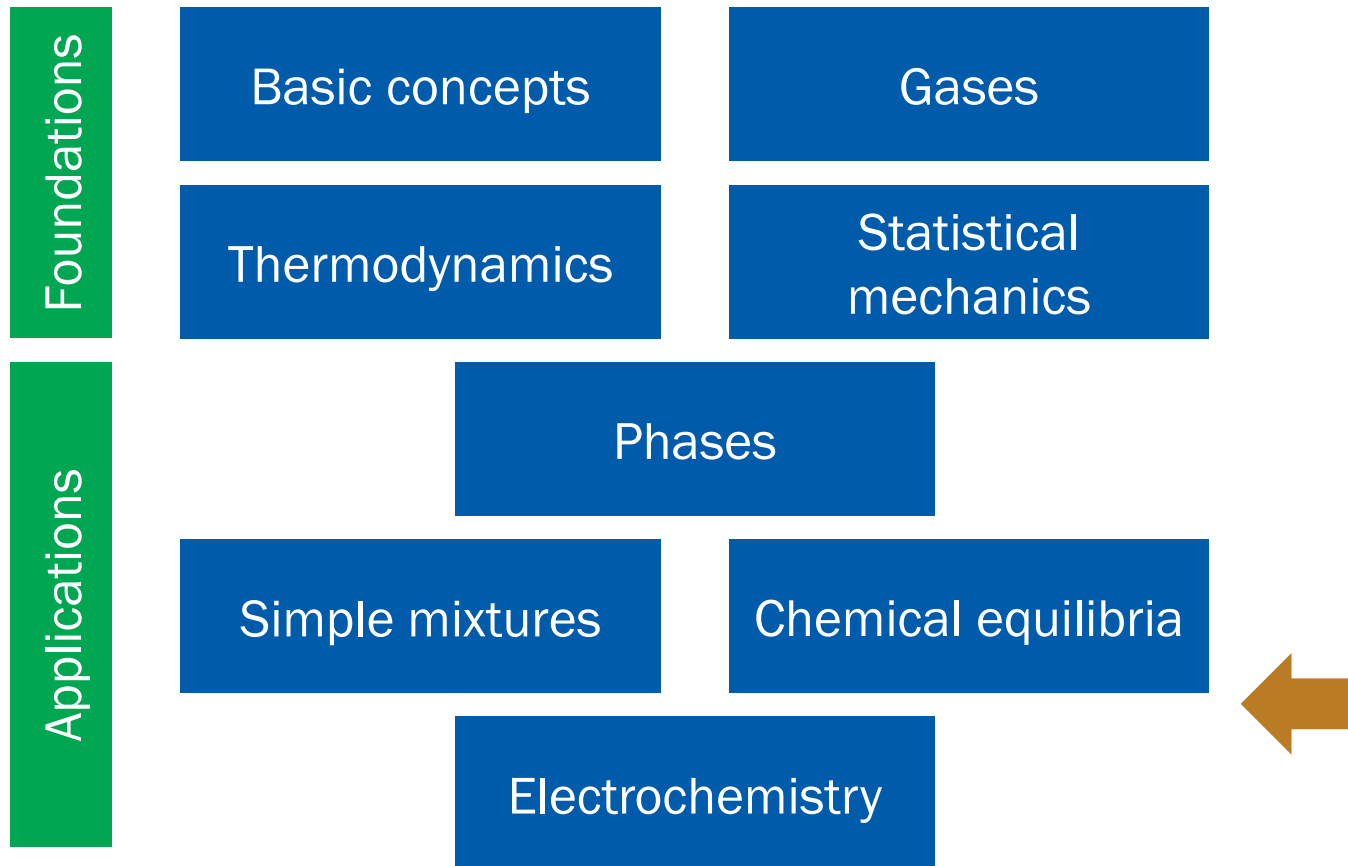
$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\ominus}{RT^2}$$

Integration leads to

$$\ln K_2 - \ln K_1 = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



Where Are We?



10. Electrochemistry

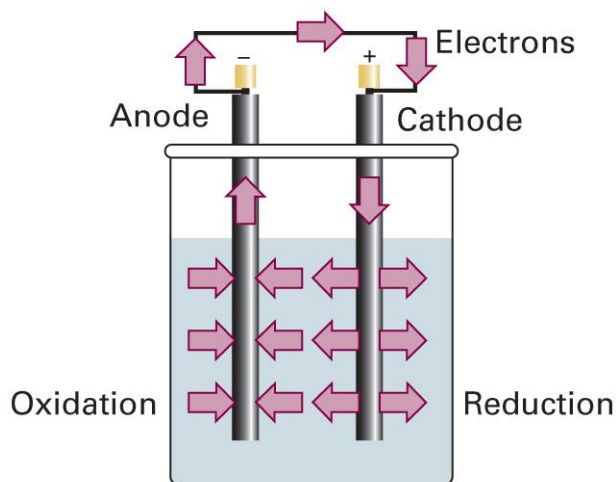
Chapter Overview

- **Electrolytes**
 - Review: electrostatics
 - Solvation energy of ions
 - Distribution of ions
 - Activities of ions
- **Electrochemical cells**
 - Electrochemical reactions
 - Thermodynamics of cells
 - Determination of ion activities



Electrochemical Cell

- Two electrodes + electrolytes
 - Anode: oxidation
 - Cathode: reduction



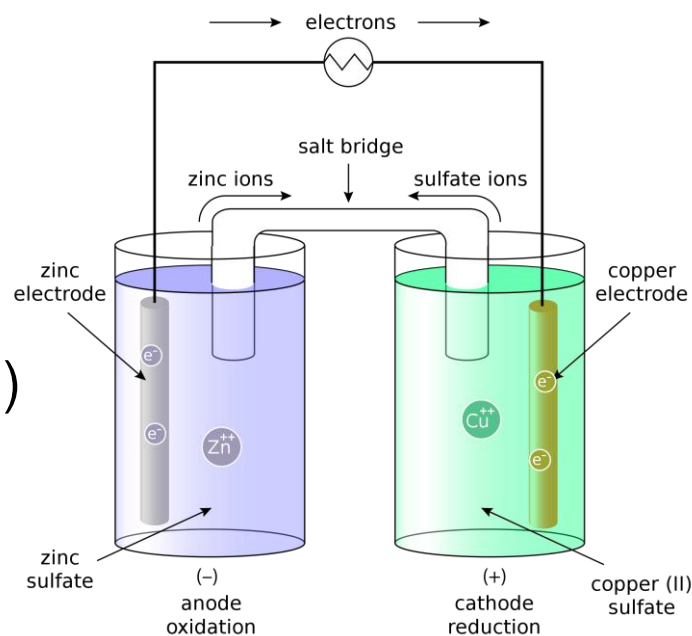
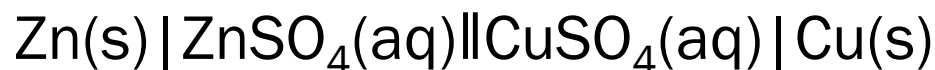
Half Cells and Cell Diagram

- Cell with $\text{Zn}^{2+}(\text{aq})/\text{Zn}(\text{s})$ and $\text{Cu}^{2+}(\text{aq})/\text{Cu}(\text{s})$ electrodes

anode: $\text{Zn}(\text{s}) \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^{-}$

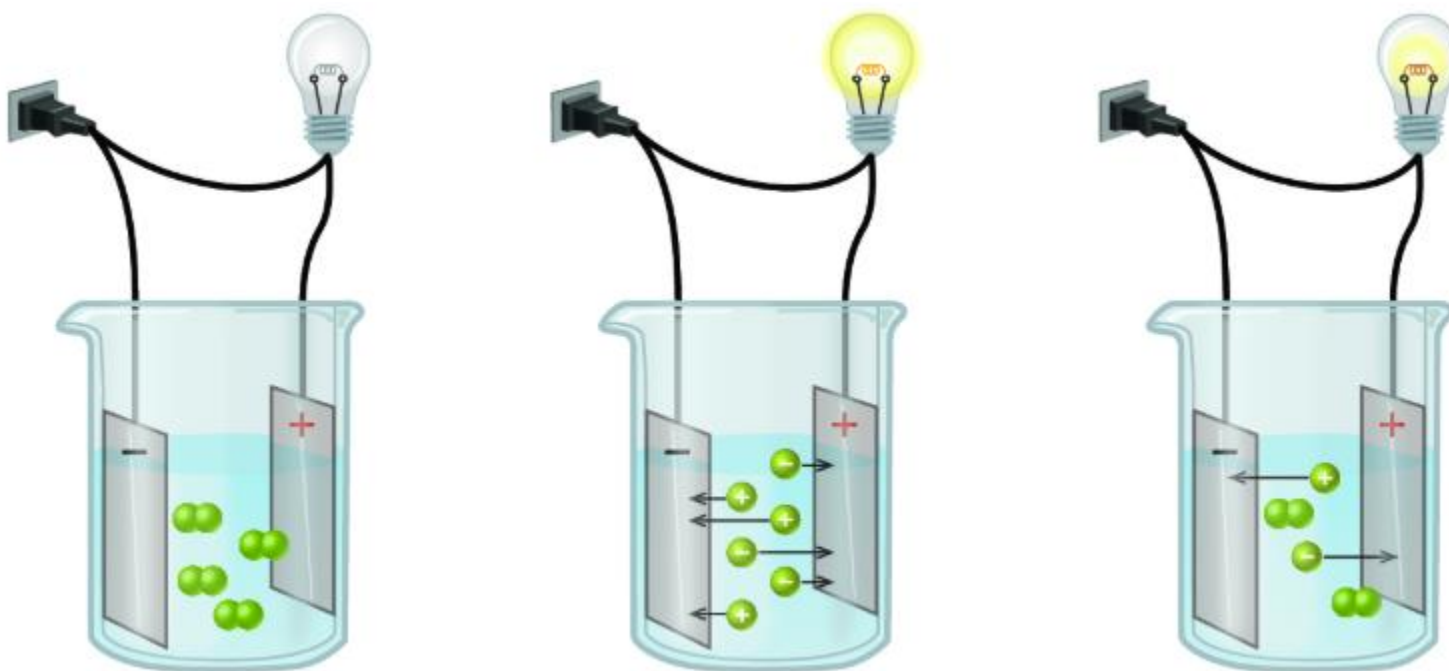
cathode: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu}(\text{s})$

- Cell diagram



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Cells and Electrolytes

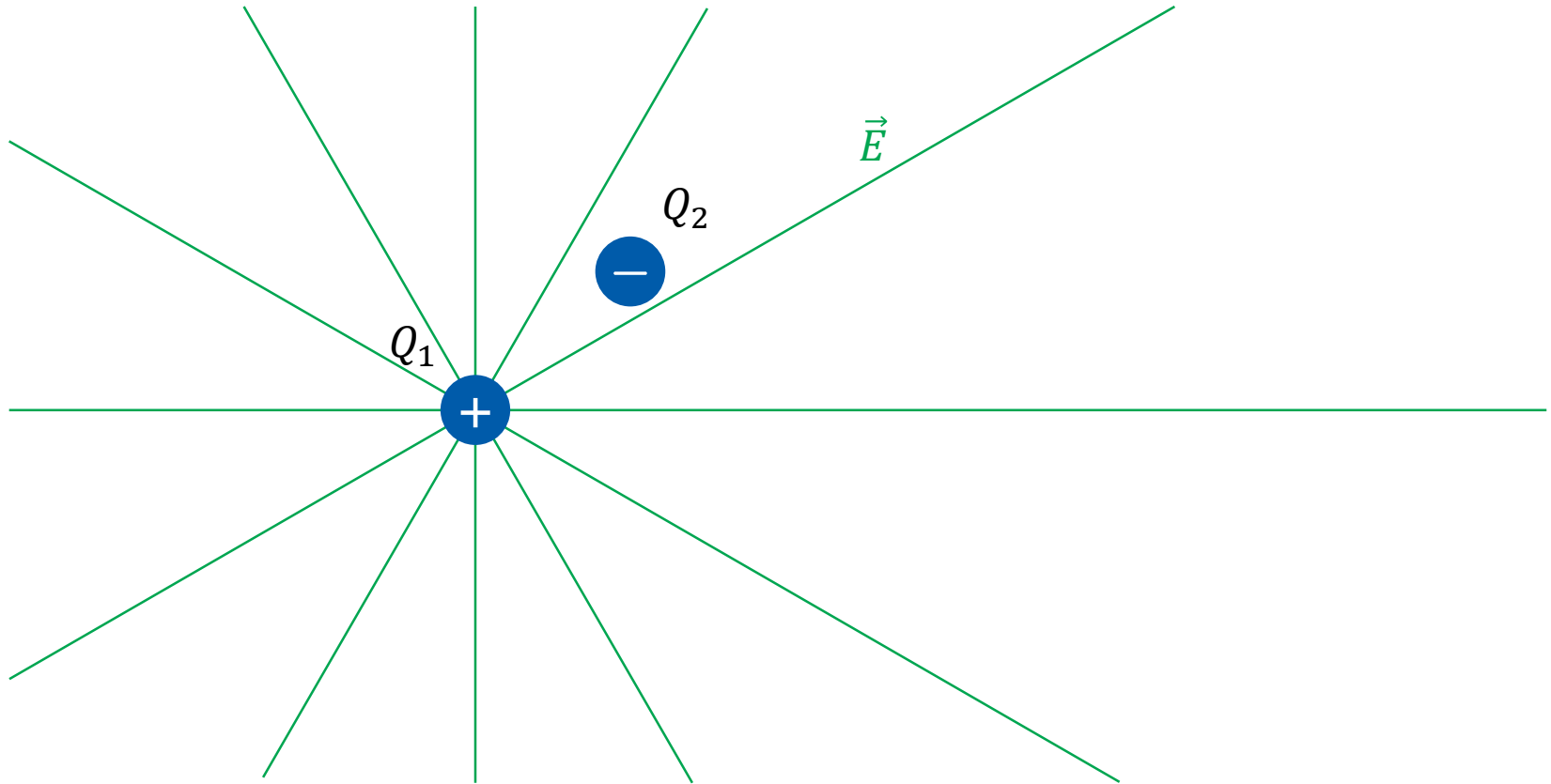


Electrostatic Force

$$\vec{F}(\vec{r}) = \frac{Q_1 Q_2}{4\pi\epsilon r^2} \hat{r}$$

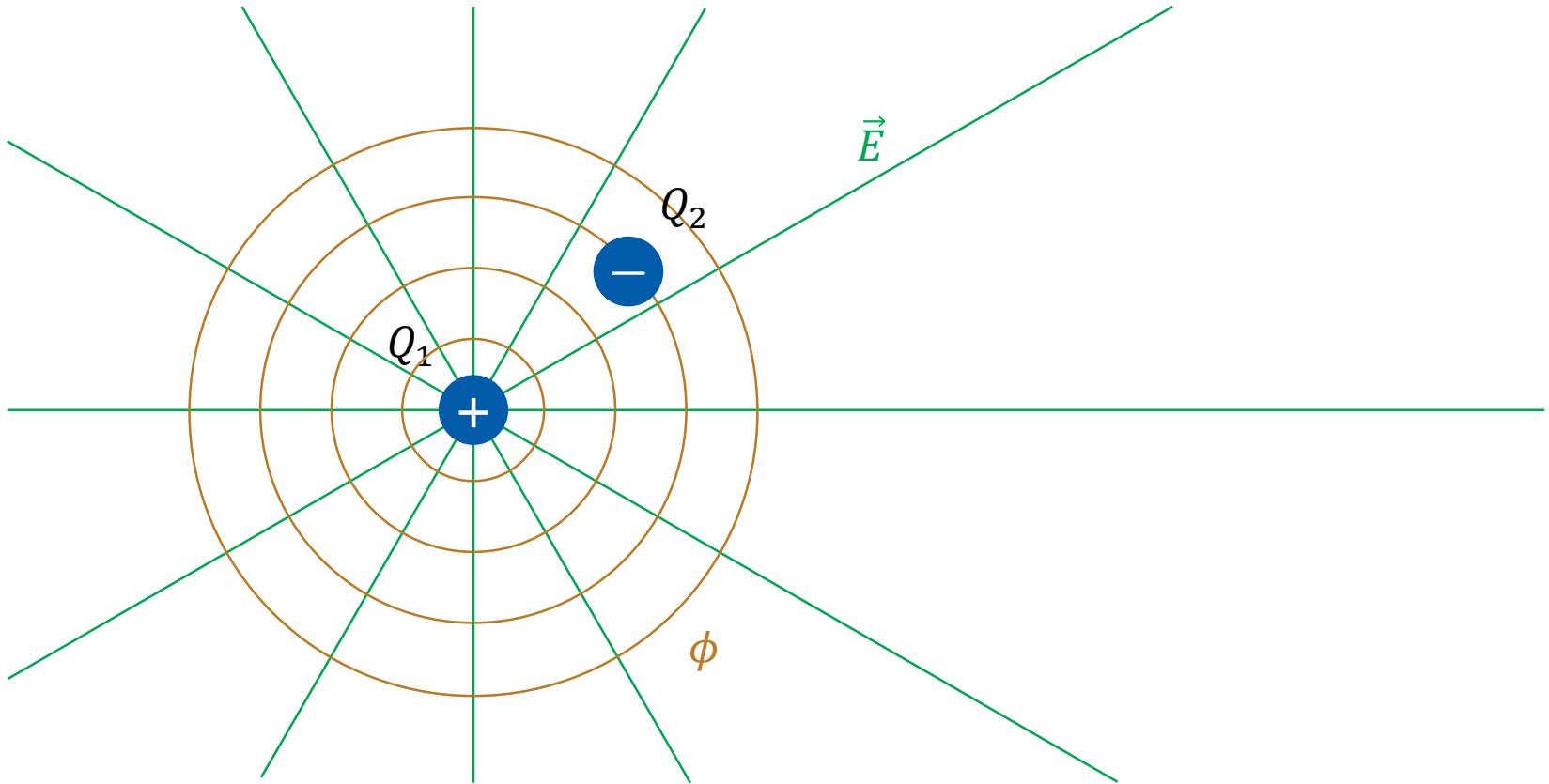


Electric Field



Review

Electric Potential



Summary: Electrostatics

- Force (Coulomb's law)

$$\vec{F}(\vec{r}) = \frac{Q_1 Q_2}{4\pi\epsilon r^2} \hat{r}$$

- Potential energy

$$E_p(r) = \frac{Q_1 Q_2}{4\pi\epsilon r}$$

- Electric field around Q_1

$$\vec{E} = \frac{Q_1}{4\pi\epsilon r^2} \hat{r}$$

$$\vec{F} = Q_2 \vec{E}$$

- Potential around Q_1

$$\phi(r) = \frac{Q_1}{4\pi\epsilon r}$$

$$E_p = Q_2 \phi$$

Poisson's Equation

Relationship between charge density and potential

For an arbitrary charge density $\rho(\vec{r})$, the potential $\phi(\vec{r})$ satisfies

$$\nabla^2 \phi = -\frac{\rho}{\epsilon}$$

$$(\text{Laplacian } \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2})$$

Electric Work

For a charge Q in the electric potential ϕ ,

$$E_p = Q\phi$$

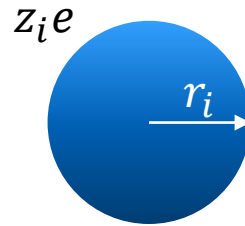
Consider its infinitesimal (reversible) change:

$$\delta w_{\text{elec}} = dE_p = d(Q\phi) = Qd\phi + \phi dQ$$

- If ϕ is the independent variable, $\delta w_{\text{elec}} = Qd\phi$
- If Q is the independent variable, $\delta w_{\text{elec}} = \phi dQ$

Work for Charging an Ion

Model each ion i as a sphere with radius r_i and charge $z_i e$.



The electric work required to charge this ion is

$$\delta w_{\text{elec}} = \phi dQ$$

$$w_{\text{elec}} = \int_0^{z_i e} \phi dQ = \int_0^{z_i e} \left(\frac{Q}{4\pi\epsilon r_i} \right) dQ = \frac{z_i^2 e^2}{8\pi\epsilon r_i}$$

Gibbs Energy for Solvation

$$w_{\text{elec}} = \frac{z_i^2 e^2}{8\pi\epsilon r_i}$$

Since $\Delta G = w_{\text{add,max}} = w_{\text{elec,max}}$,

$$\Delta_{\text{solv}} G^\ominus \approx w(\text{medium}) - w(\text{vacuum})$$

For vacuum, $\epsilon = \epsilon_0$ and

$$\Delta_{\text{solv}} G^\ominus = \frac{z_i^2 e^2}{8\pi\epsilon r_i} - \frac{z_i^2 e^2}{8\pi\epsilon_0 r_i} = -\frac{z_i^2 e^2}{8\pi\epsilon_0 r_i} \left(1 - \frac{\epsilon_0}{\epsilon}\right)$$

Born Equation

$$\Delta_{\text{solv}} G^{\ominus} = -\frac{z_i^2 e^2}{8\pi\epsilon_0 r_i} \left(1 - \frac{1}{\epsilon_r}\right)$$

where the relative permittivity $\epsilon_r = \epsilon/\epsilon_0$.

- $\Delta_{\text{solv}} G^{\ominus} < 0$: solvation of ions is favored
- Water has $\epsilon_r = 78.54$ at 25°C, and

$$\Delta_{\text{solv}} G^{\ominus} = -\frac{z_i^2}{r_i/\text{pm}} \times (6.86 \times 10^4 \text{ kJ mol}^{-1})$$

Generalized Born Model

$$\Delta_{\text{solv}} G^\ominus = -\frac{z_i^2 e^2}{8\pi\epsilon_0} \left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{r_i}$$

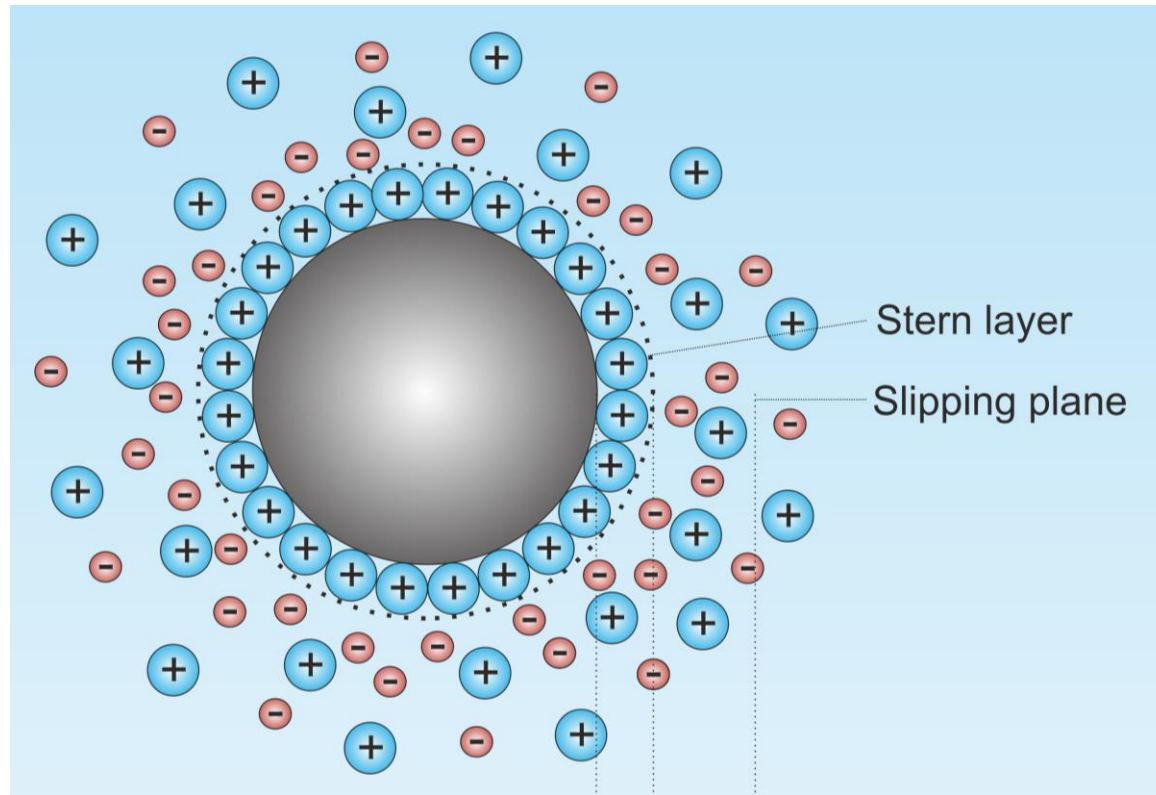
We can generalize the Born model by using

$$\Delta_{\text{solv}} G^\ominus = -\frac{z_i^2 e^2}{8\pi\epsilon_0} \left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{\sqrt{r_i^2 + a_i^2 e^{-(r_i/2a_i)^2}}}$$

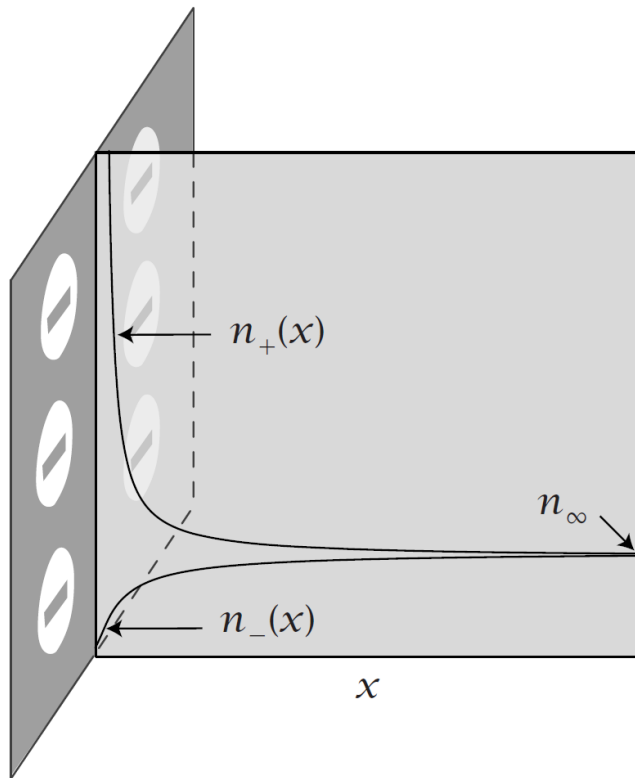
where a_i is the effective Born radius.

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Distribution of Ions in ϕ



Boltzmann Distribution



In the electric potential ϕ ,

- Potential energy

$$E_p = Q\phi$$

- Boltzmann distribution

$$P \propto e^{-E_p/k_B T} = e^{-Q\phi/k_B T}$$

- Mole numbers of ions

$$n_+ = n_{\infty} e^{-ze\phi/k_B T}$$

$$n_- = n_{\infty} e^{+ze\phi/k_B T}$$

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Poisson-Boltzmann Equation

- Charge density

$$\begin{aligned}\rho &= zeN_A n_+ / V + (-ze)N_A n_- / V = \overset{\text{Faraday constant } eN_A}{zF}(n_+ - n_-) / V \\ &= zF n_\infty (e^{-ze\phi/k_B T} - e^{+ze\phi/k_B T}) / V\end{aligned}$$

- Poisson equation

$$\nabla^2 \phi = -\rho / \epsilon$$

- Poisson-Boltzmann equation

$$\nabla^2 \phi = \frac{zF n_\infty}{\epsilon V} (e^{+ze\phi/k_B T} - e^{-ze\phi/k_B T})$$

Debye-Hückel Equation

$$\nabla^2 \phi = \frac{zF n_{\infty}}{\epsilon V} \left(e^{+ze\phi/k_B T} - e^{-ze\phi/k_B T} \right)$$

At the limit of a dilute solution, $ze\phi/k_B T \ll 1$ and

$$e^{\pm ze\phi/k_B T} \approx 1 \pm ze\phi/k_B T = 1 \pm zF\phi/RT$$

Hence,

$$\nabla^2 \phi = \frac{zF n_{\infty}}{\epsilon V} \{ (1 + zF\phi/RT) - (1 - zF\phi/RT) \}$$

$$\nabla^2 \phi = \frac{2z^2 F^2 n_{\infty}}{\epsilon V R T} \phi$$

Ion Strength

$$I = \frac{1}{2} \sum_i z_i^2 \left(\frac{b_i}{b^\ominus} \right)$$

For two types of ions with $|z_+| = |z_-| = z$,

$$I = \frac{z_+^2 b_+ + z_-^2 b_-}{2b^\ominus} = \frac{z^2 b}{b^\ominus}$$

Since $b = n_B/m_A = n_\infty/m_A$,

$$I = \frac{z^2 n_\infty}{b^\ominus m_A}$$

Generalization of D-H Eq.

$$\nabla^2 \phi = \frac{2F^2 z^2 n_{\infty}}{\varepsilon V R T} \phi$$

Since the ion strength is $I = z^2 n_{\infty} / b^{\ominus} m_A$,

$$\nabla^2 \phi = \frac{2F^2 I b^{\ominus} (m_A / V)}{\varepsilon R T} \phi$$

This equation is generally applicable to any salt system.

Debye Length

$$r_D = \sqrt{\frac{\varepsilon RT}{2F^2 I b^\ominus(m_A/V)}} \propto \frac{1}{\sqrt{I}}$$

Then

$$\nabla^2 \phi = \phi / r_D^2$$

For an aqueous solution at room temperature,

$$r_D \approx (0.304 \text{ nm M}^{1/2}) / \sqrt{I}$$

Screening Effect

For a point charge of Q , solving the differential equation

$$\nabla^2 \phi = \phi / r_D^2$$

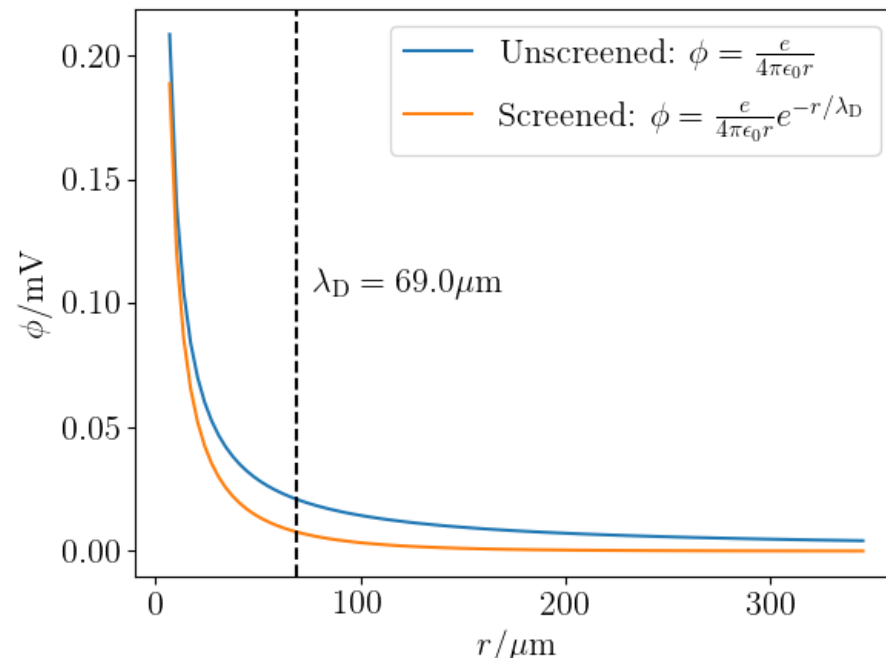
gives a solution of

$$\phi(r) = \frac{Q}{4\pi\epsilon r} e^{-r/r_D}$$

which can be compared to the Coulombic potential

$$\phi(r) = \frac{Q}{4\pi\epsilon r}$$

Screening Effect



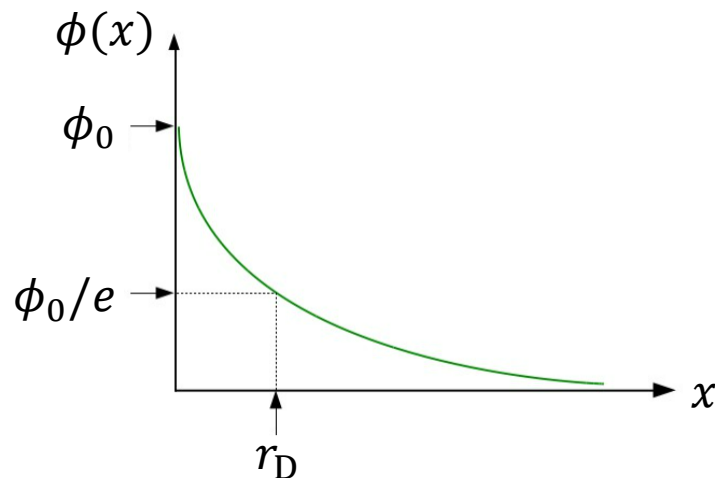
Ions Around Electrodes

For a charged plate, solving the differential equation

$$\nabla^2 \phi = \phi / r_D^2$$

gives a solution of

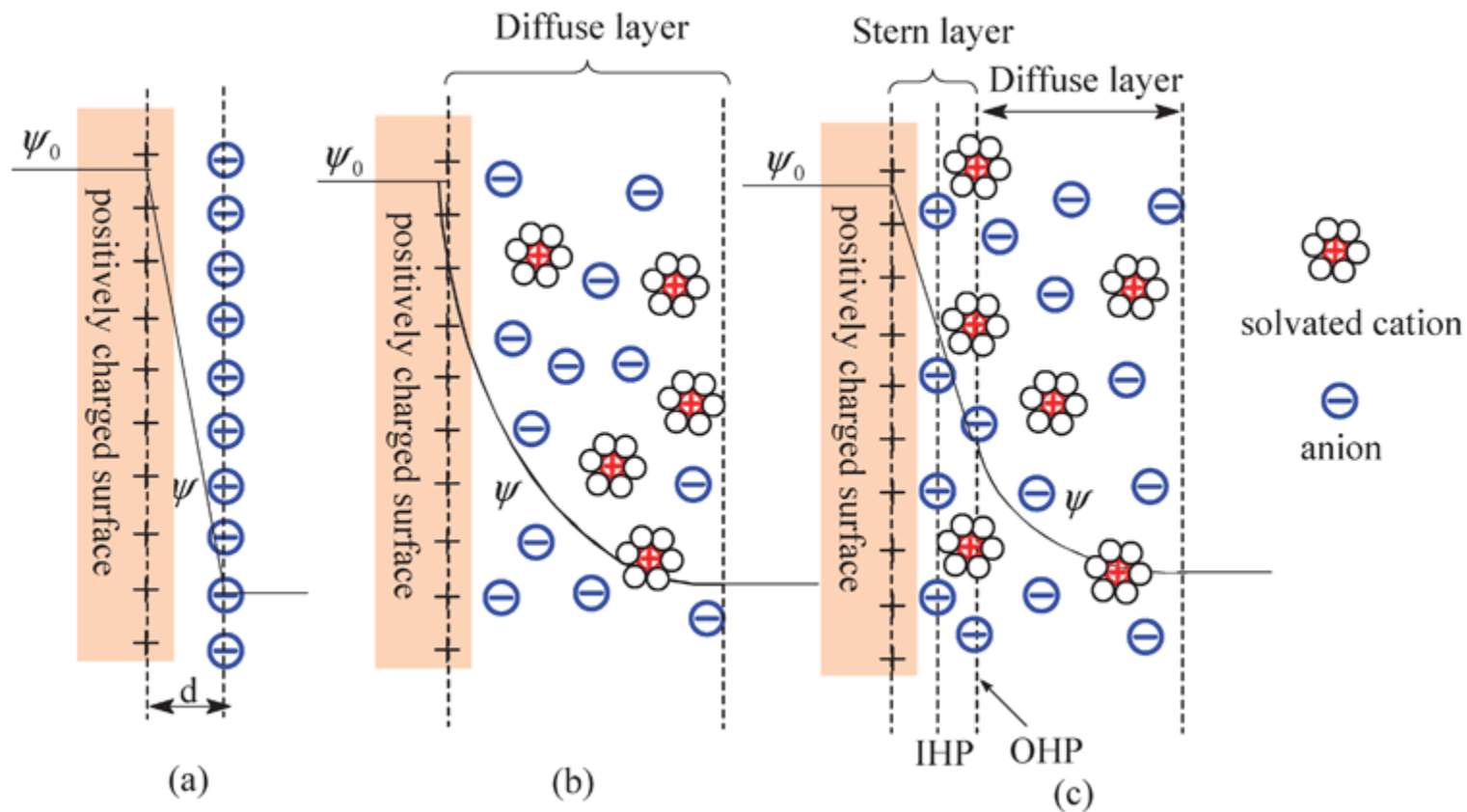
$$\phi(x) = \phi_0 e^{-x/r_D}$$



(Image: <https://docplayer.net/21695464-Charged-interfaces-formation-of-charged-surfaces-helmholz-gouy-chapman-and-stern-model-of-double-layer.html>)

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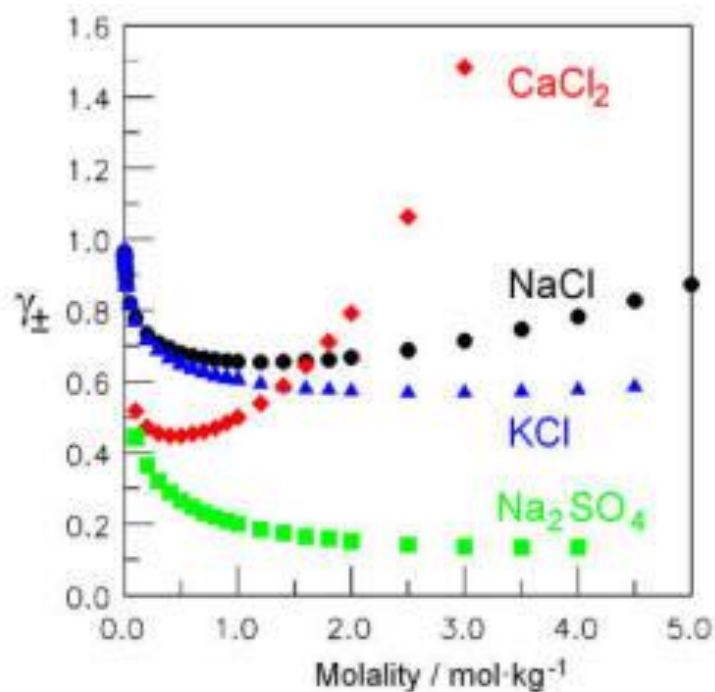
Electric Double Layer (EDL)



(Image: Zhang and Zhao, *Chemical Society Reviews* 38: 2520-2531 (2009), Figure 2)

Activities of Ions

Quickly deviate from ideal behavior



(Image: <https://chem.libretexts.org/@go/page/44345>)

Molar Gibbs Energy of Ions

For a real solution consisting of M^+ and X^- ,

$$G_m = \mu_+ + \mu_-$$

However,

$$\mu_J = \mu_J^\ominus + RT \ln a_J = \mu_J^\ominus + RT \ln(\gamma_J x_J) = \mu_J^{\text{ideal}} + RT \ln \gamma_J$$

and

$$G_m = \mu_+ + \mu_- = \mu_+^{\text{ideal}} + \mu_-^{\text{ideal}} + RT \ln \gamma_+ + RT \ln \gamma_-$$

Hence,

$$G_m = G_m^{\text{ideal}} + RT \ln(\gamma_+ \gamma_-)$$

Mean Activity Coefficients

$$G_m = G_m^{\text{ideal}} + RT \ln(\gamma_+ \gamma_-)$$

Experimentally impossible to separate γ_+ and γ_-

- **Mean activity coefficient:** geometric mean of γ_+ and γ_-

$$\gamma_{\pm} = (\gamma_+ \gamma_-)^{1/2}$$

Hence,

$$\mu_+ = \mu_+^{\text{ideal}} + RT \ln \gamma_{\pm}$$

$$\mu_- = \mu_-^{\text{ideal}} + RT \ln \gamma_{\pm}$$

Generalization

For a salt M_pX_q ($= pM^{|z_+|+} + qX^{|z_-|-}$),

$$G_m = p\mu_+ + q\mu_- = G_m^{\text{ideal}} + pRT \ln \gamma_+ + qRT \ln \gamma_-$$

The mean activity coefficient is defined as

$$\gamma_{\pm} = (\gamma_+^p \gamma_-^q)^{1/(p+q)}$$

so that

$$\mu_J = \mu_J^{\text{ideal}} + RT \ln \gamma_{\pm}$$

Activity and Work

$$\gamma_{\pm} = (\gamma_+^p \gamma_-^q)^{1/(p+q)}$$

Since $G_m = G_m^{\text{ideal}} + pRT \ln \gamma_+ + qRT \ln \gamma_-$,

$$G_m = G_m^{\text{ideal}} + RT \ln(\gamma_+^p \gamma_-^q) = G_m^{\text{ideal}} + (p + q)RT \ln \gamma_{\pm}$$

Hence,

$$\ln \gamma_{\pm} = \frac{G_m - G_m^{\text{ideal}}}{(p + q)RT} = \frac{\overset{\text{work of charging an ion}}{w_e}}{(p + q)RT}$$

Work of Charging an Ion

- The work of charging an ion:

$$\delta w_e = \phi_{\text{others}} dQ$$

- The potential due to other ions:

$$\phi_{\text{others}} = \phi_{\text{DH}} - \phi_{\text{self}} = \frac{Q}{4\pi\epsilon r} e^{-r/r_D} - \frac{Q}{4\pi\epsilon r}$$

As we are interested in the vicinity of the ion ($r \rightarrow 0$),

$$e^{-r/r_D} \approx 1 - r/r_D$$

$$\delta w_e = -\frac{Q}{4\pi\epsilon r_D} dQ$$

Work of Charging an Ion

$$\delta w_e = -\frac{Q}{4\pi\epsilon r_D} dQ$$

For an ion of charge $z_i e$,

$$w_e = -\frac{1}{4\pi\epsilon r_D} \int_0^{z_i e} Q dQ = -\frac{z_i^2 e^2}{8\pi\epsilon r_D}$$

The molar work is

$$w_{e,m} = -\frac{N_A z_i^2 e^2}{8\pi\epsilon r_D} = -\frac{z_i^2 F^2}{8\pi N_A \epsilon r_D}$$

Total Work

$$w_{e,m} = -\frac{z_i^2 F^2}{8\pi N_A \epsilon r_D}$$

The total work of charging p moles of M and q moles of X:

$$w_{e,\text{total}} = pw_{e,m}(\text{M}) + qw_{e,m}(\text{X}) = -\frac{(pz_+^2 + qz_-^2)F^2}{8\pi N_A \epsilon r_D}$$

Since $pz_+ + qz_- = 0$ (charge neutrality),

$$pz_+^2 + qz_-^2 = z_+(-qz_-) + z_-(-pz_+) = -(p+q)z_+z_-$$

$$w_{e,\text{total}} = \frac{(p+q)z_+z_-F^2}{8\pi N_A \epsilon r_D} = -\frac{(p+q)|z_+z_-|F^2}{8\pi N_A \epsilon r_D}$$

Activity

$$w_{\text{e,total}} = - \frac{(p + q)|z_+ z_-| F^2}{8\pi N_A \epsilon r_D}$$

$$\ln \gamma_{\pm} = \frac{w_e}{(p + q)RT} = - \frac{|z_+ z_-| F^2}{8\pi N_A RT \epsilon r_D}$$

Since $r_D = \{\epsilon RT / 2F^2 I b^{\ominus}(m_A/V)\}^{1/2}$,

$$\ln \gamma_{\pm} = -|z_+ z_-| \left[\frac{F^3}{4\pi N_A} \left\{ \frac{b^{\ominus}(m_A/V)}{2\epsilon^3 R^3 T^3} \right\}^{1/2} \right] I^{1/2}$$

Debye-Hückel Limiting Law

$$\ln \gamma_{\pm} = -|z_+ z_-| \left[\frac{F^3}{4\pi N_A} \left\{ \frac{b^{\ominus}(m_A/V)}{2\varepsilon^3 R^3 T^3} \right\}^{1/2} \right] I^{1/2}$$

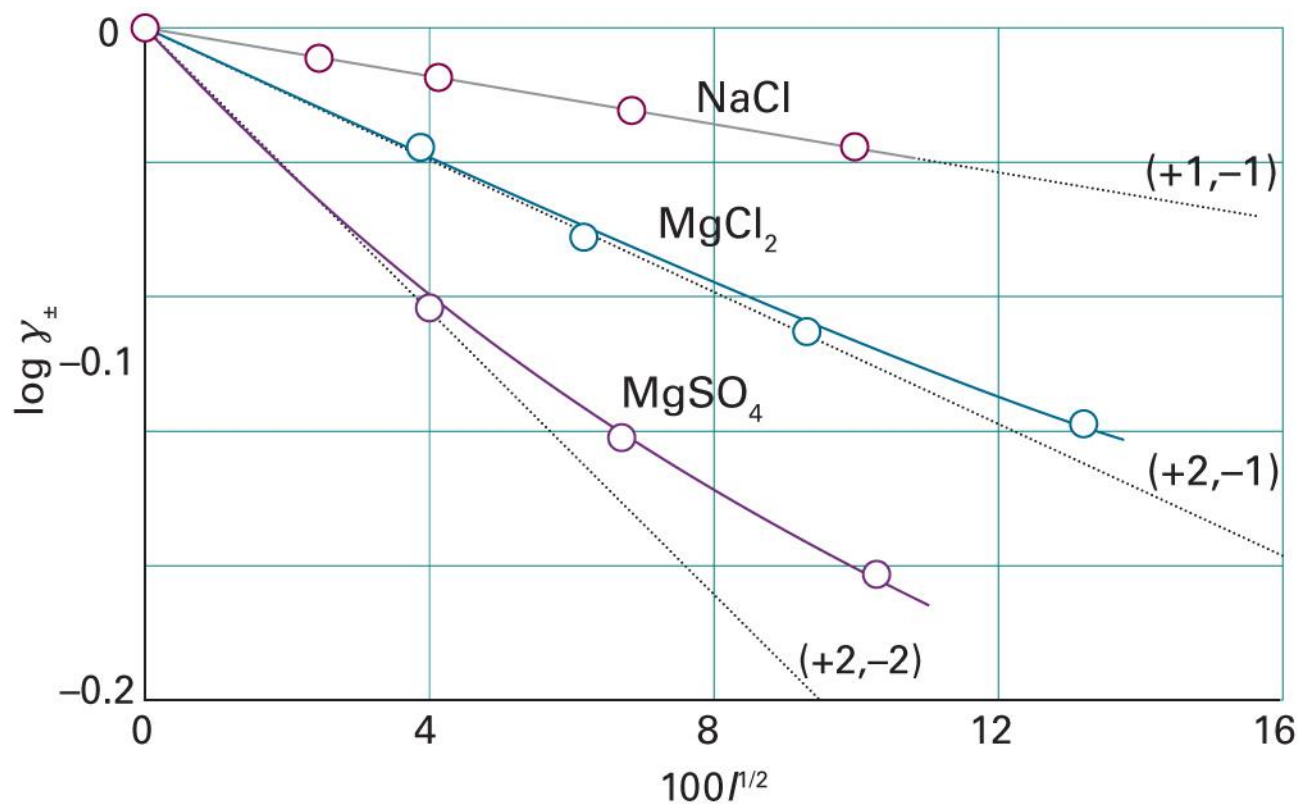
$$\log_{10} \gamma_{\pm} = -|z_+ z_-| \left[\frac{F^3}{4\pi N_A \ln 10} \left\{ \frac{b^{\ominus}(m_A/V)}{2\varepsilon^3 R^3 T^3} \right\}^{1/2} \right] I^{1/2}$$

Hence,

$$\log_{10} \gamma_{\pm} = -A|z_+ z_-| I^{1/2}$$

where $A = 0.509$ for an aqueous solution at 25°C

Debye-Hückel Limiting Law



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5F.5)

Extensions

- Debye-Hückel law (1923)

$$\log_{10} \gamma_{\pm} = -A|z_+z_-|I^{1/2}$$

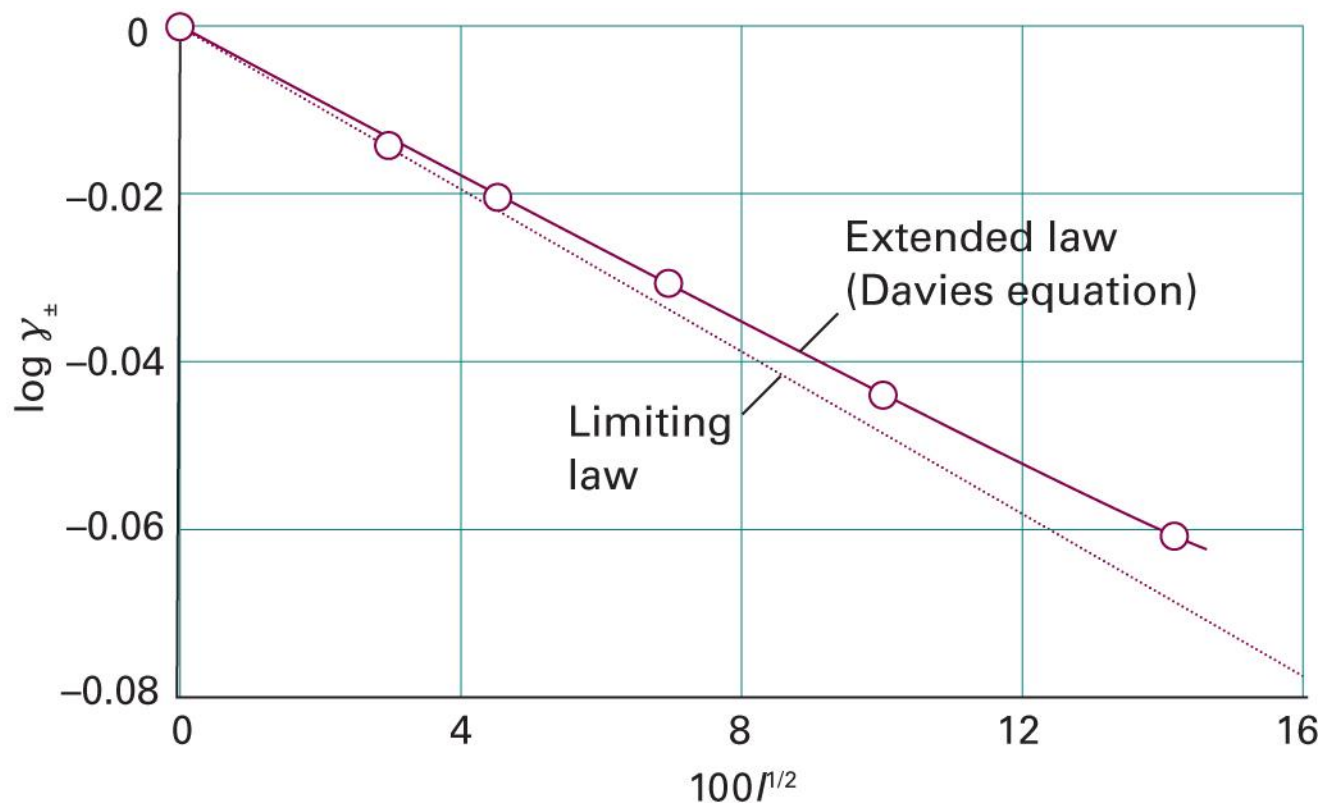
- Extended Debye-Hückel law

$$\log_{10} \gamma_{\pm} = -\frac{A|z_+z_-|I^{1/2}}{1 + BI^{1/2}}$$

- Davies equation (1938)

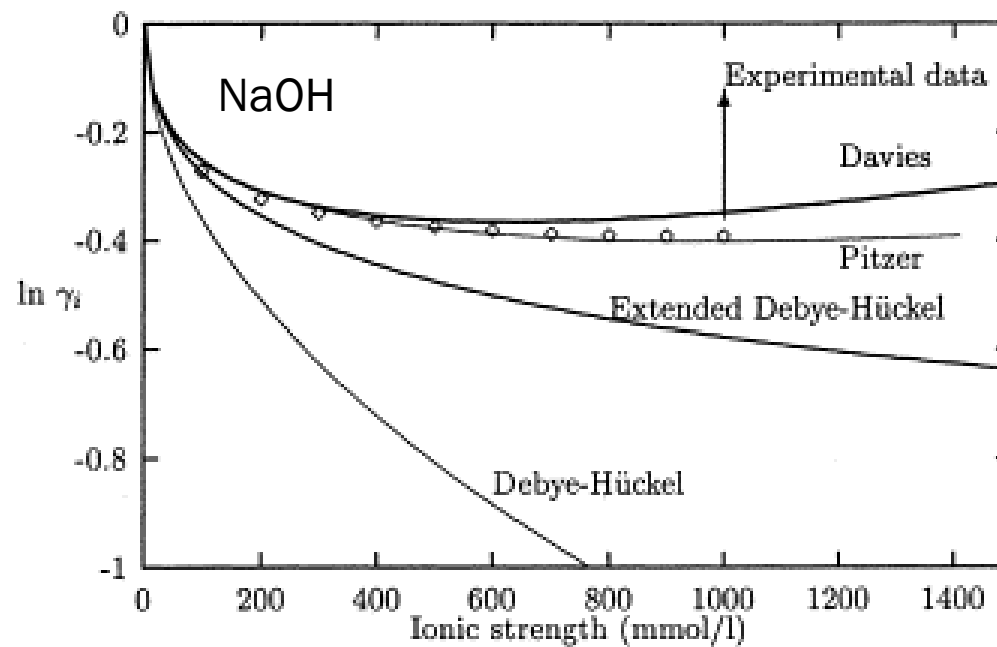
$$\log_{10} \gamma_{\pm} = -\frac{A|z_+z_-|I^{1/2}}{1 + BI^{1/2}} + CI$$

Model Performances



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5F.6)

At Higher Concentrations



(Image: Samson et al., *Computational Materials Science* 15: 285-294 (1999), Figure 1)